

Machine Learning Model Evidence Pack

Section: Model Predictions

Scope of this evidence

The evidence section collates the outputs produced by the trained Random Forest model. It includes the notebook's model-performance metric outputs, probability diagnostics, and the final student-level attrition predictions persisted for application use.

1. Model Performance Metric Outputs

The notebook evaluates the fitted Random Forest on both the validation and untouched test splits. Recall is identified as the primary metric, with accuracy, precision, F1, ROC-AUC and mean squared error (MSE) reported alongside confusion-matrix counts and predicted-class totals.

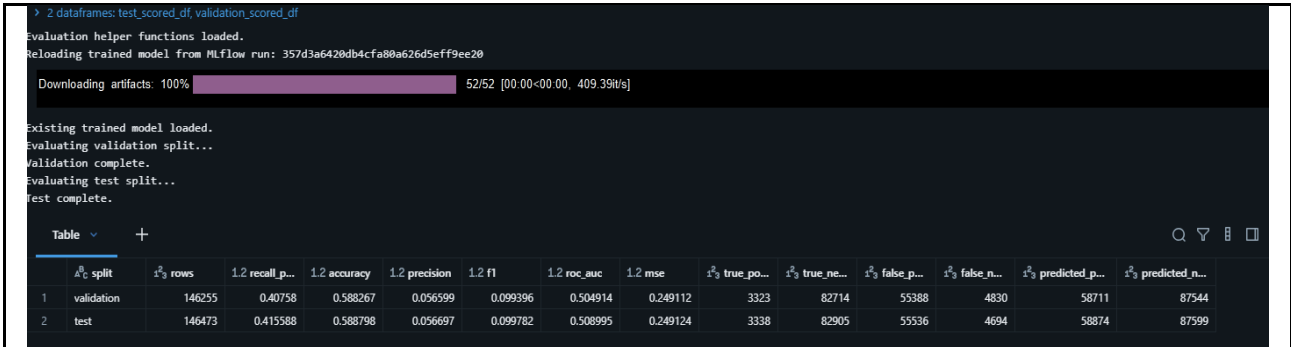


Figure 1. Validation and test performance metrics produced by the trained Random Forest model (notebook Section 21, cell 71).

Metric	Validation	Test	Meaning
Rows	146,255	146,473	Number of records evaluated in the split.
Recall (primary)	0.40758	0.415588	Proportion of actual attrition cases correctly identified by the model.
Accuracy	0.588267	0.588798	Proportion of all evaluated records classified correctly.
Precision	0.056599	0.056697	Proportion of predicted attrition cases that are actual attrition cases.
F1	0.099396	0.099782	Harmonic mean of precision and recall.
ROC-AUC	0.504914	0.508995	Threshold-independent measure of the model's ability to rank positive cases above negative cases.
MSE	0.249112	0.249124	Mean squared error between the actual label and the model's continuous positive-class probability.

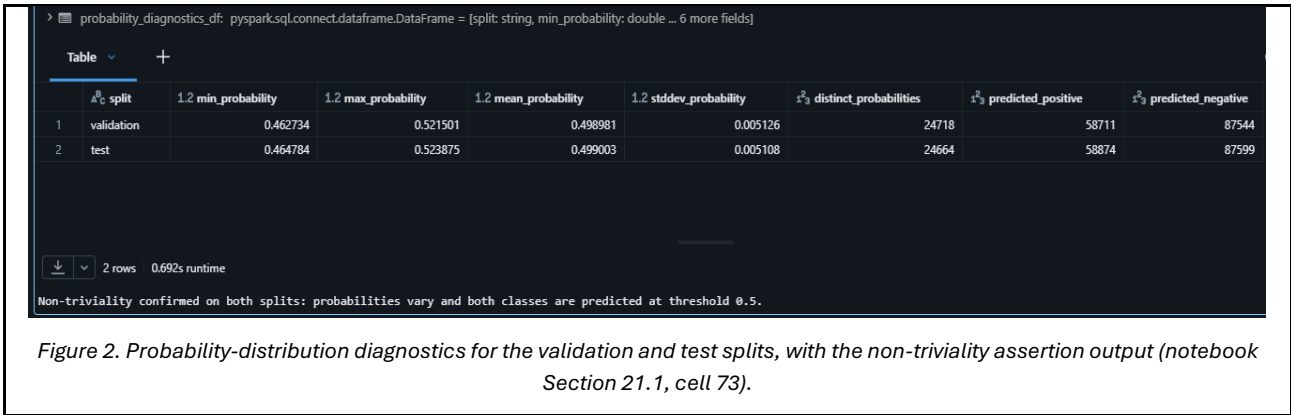
Metric	Validation	Test	Meaning
True positives (TP)	3,323	3,338	Actual attrition cases correctly predicted as positive.
True negatives (TN)	82,714	82,905	Actual non-attrition cases correctly predicted as negative.
False positives (FP)	55,388	55,536	Non-attrition cases incorrectly predicted as at risk.
False negatives (FN)	4,830	4,694	Attrition cases incorrectly predicted as not at risk.
Predicted positive	58,711	58,874	Total records classified as positive at the 0.50 threshold.
Predicted negative	87,544	87,599	Total records classified as negative at the 0.50 threshold.

Notebook source

Section 21 displays these columns: split, rows, recall_primary, accuracy, precision, f1, roc_auc, mse, true_positives, true_negatives, false_positives, false_negatives, predicted_positive and predicted_negative.

2. Probability Distribution Outputs

The notebook also reports probability-distribution diagnostics for the validation and test splits. These outputs provide evidence that the model generates varying risk scores and predicts both classes rather than returning a constant score or a single class.



Output	Validation	Test	Meaning
Minimum probability	0.462734	0.464784	Lowest positive-class probability produced.
Maximum probability	0.521501	0.523875	Highest positive-class probability produced.
Mean probability	0.498981	0.499003	Average positive-class probability.
Probability standard deviation	0.005126	0.005108	Spread of the positive-class probability values.
Distinct probabilities	24,718	24,664	Number of distinct probability scores after rounding to six decimal places.
Predicted positive	58,711	58,874	Count classified as positive at threshold 0.50.
Predicted negative	87,544	87,599	Count classified as negative at threshold 0.50.

Non-triviality check

The notebook explicitly asserts that probabilities are not constant, probability standard deviation is greater than zero, and both positive and negative predictions occur at the 0.50 threshold.

3. Student Attrition Risk Predictions

The trained Random Forest model scores applicable student records and produces an attrition-risk result for each student. The final prediction set is persisted to the approved Delta table so downstream application components can retrieve a prediction using the student’s de-identified identifier.

```
display(spark.table("workspace.student_aggregate.student_attrition_risk_prediction").orderBy(F.rand()).limit(20))
```

	student_deidentified_hash	attrition_risk_percentage	attrition_risk_flag	prediction_threshold	mlflow_run_id	scored_at
1	41d723482924943fceff69fa2560dfc4187e7fc515364bbd0a9e6172dd5dae7b	49.2414510855733	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
2	2650bdf90cc4435f4983759ccab15f9825dea3f588f6c7ea99812aa87a45719a	48.95807292635754	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
3	> 377d2ef3e9141fd05de313beec003ab5a1e74061d9a03450118f2aad2...	50.47389403354223	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
4	c3511b87e6d22be2bfa704b7dd7175245a01fa5f073a4a3e74ed54624c935	50.50296951955872	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
5	> d2640e350c99d746d77e171d5b6a055ef8bffa161e42616bbe522e05145...	50.392419165010985	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
6	> 63ae6c6913b4bd7075e5f1165bb92e5d6e957f54d97e64d8a27b0661...	49.63105646408955	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
7	> b83f04aaa51ce7d4b759c1c74183f981077e35a643ac5b59dfe3ce803b...	49.78247644646776	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
8	84b4f7e3a2186db29695580f6d5f5be3c192b5f56a1800a0c4dfe6ec2b82fa21	49.55091108290457	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
9	1c9c9f6c995a222e99337de2997f8b5fffe51f8e6c09813286cf8aed64e15e	50.3147621711729	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
10	> 5ba1e8c6b561c3e456b51e9645a2abe2d784586a4a656300eb241addb...	49.6033113021228	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
11	29c095f07f78281f9b72ada1632528c4fc49be45f6b7a6e6c695f8d9c7a4b2f	51.70406340256446	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
12	> f0b387e7b47b88ac3391836488009be5552b2d00aeb680b7725d06b67c...	49.798532135578164	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
13	> 083ced43b46b9a31cd2294152da3455cd32514b2903cb8486362b69b5...	50.3991505754436	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
14	> 79c007b9722169747d481bd1711b882034056ee79a0fa1c0a2ca27639...	50.449275254745714	true	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...
15	6589efTee5a370284f8f035a3418219aadcb7a04425b9a18f95bc5f99aae64a	49.54079922865124	false	0.5	357d3a6420db4cfa80a626d5eff9ee...	2026-08-06T15:46:31.493+00:...

Figure 3. Random sample of persisted records from workspace.student_aggregate.student_attrition_risk_prediction (notebook Section 30, cell 108).

Prediction output fields

Field	Purpose
student_deidentified_hash	Logical student identifier used to retrieve a prediction without exposing a direct student identifier.
attrition_risk_percentage	Model-produced relative attrition-risk score expressed on a 0–100 scale.
attrition_risk_flag	Boolean application flag indicating whether the model score meets the approved risk threshold.
prediction_threshold	Stores the threshold used when assigning the Boolean risk flag. The approved threshold is 0.50.
mlflow_run_id	Links each persisted prediction to the MLflow model run that produced it.
scored_at	Timestamp recording when the prediction set was generated.

4. Prediction Logic and Persistence

The model first generates an unrounded positive-class probability. The approved decision rule converts that score into the stored Boolean attrition-risk flag:

Risk classification rule

attrition_risk_flag = TRUE when the unrounded model score is ≥ 0.50 .
attrition_risk_flag = FALSE when the unrounded model score is < 0.50 .

The final prediction set is written to:

```
workspace.student_aggregate.student_attrition_risk_prediction
```

The notebook records this as the approved six-column persisted schema. The intermediate raw probability is used during staging and checking but is not added as a seventh column to the final published table.

Reproducible evidence query

```
display(
  spark.table("workspace.student_aggregate.student_attrition_risk_prediction")
    .orderBy(F.rand())
    .limit(20)
)
```

5. Interpretation of the Prediction Output

Important interpretation

attrition_risk_percentage should be presented as a relative risk/ranking score, not as a calibrated likelihood of attrition. The notebook documents that class weighting recentres model scores around the 0.50 decision boundary. For example, a stored score of 52 should not be described to staff as a literal 52% probability that the student will attrit.

For the evidence pack, the prediction table demonstrates the operational output required by the application: a student-level risk score and a clear at-risk/not-at-risk flag that can be retrieved from Delta by the student's de-identified key. The metric and probability-diagnostic tables above provide supplementary evidence of the model outputs generated during evaluation.

6. Summarised Evidence Commentary

Summarised Evidence commentary

The trained Random Forest model was evaluated on a held-out validation split of 146,255 records and an untouched test split of 146,473 records, with recall on the positive attrition class treated as the primary metric. Recall was 0.40758 on validation and 0.415588 on test. Accuracy was 0.588267 and 0.588798, precision 0.056599 and 0.056697, F1 0.099396 and 0.099782, ROC-AUC 0.504914 and 0.508995, and MSE 0.249112 and 0.249124. The close agreement between the two splits indicates the reported performance is stable rather than an artefact of the validation split. Metrics are computed for the positive class rather than as weighted averages, because a negative class of roughly 94.5 per cent would otherwise dominate and misrepresent the primary metric.

The confusion-matrix counts confirm that both classes are predicted at the approved 0.50 threshold. On validation the model returned 3,323 true positives, 82,714 true negatives, 55,388 false positives and 4,830 false negatives, giving 58,711 predicted positive and 87,544 predicted negative. On test it returned 3,338 true positives, 82,905 true negatives, 55,536 false positives and 4,694 false negatives, giving 58,874 predicted positive and 87,599 predicted negative. The large false-positive count is the expected consequence of class weighting applied to a heavily imbalanced target, and reflects the deliberate decision to favour recall over precision for an early-warning use case.

The probability-distribution diagnostics evidence non-trivial scoring behaviour. Positive-class probabilities span 0.462734 to 0.521501 on validation and 0.464784 to 0.523875 on test, with means of 0.498981 and 0.499003, standard deviations of 0.005126 and 0.005108, and 24,718 and 24,664 distinct probability values after rounding to six decimal places. The notebook asserts that probabilities are not constant, that the standard deviation is greater than zero, and that both positive and negative predictions occur at the 0.50 threshold. That assertion passed on both splits, confirming that the model produces a varying, rankable risk score rather than a constant score or a single predicted class.

The fitted model was then applied exactly once to the approved inference population of 973,770 student rows, covering 973,770 distinct students with no unknown outcomes. Every applicable student produced one validated result and 973,770 rows were persisted. Applying the approved decision rule to the unrounded model score, 391,388 students (40.1931 per cent) were assigned attrition_risk_flag = true and 582,382 students (59.8069 per cent) were assigned false, across a probability range of 0.461106 to 0.524301 with a mean of 0.499000. Twelve pre-publish checks and nine post-publish checks passed, covering absence of nulls, uniqueness of students, probabilities within [0,1], percentages within [0,100], Boolean-only flags, a threshold of exactly 0.50, and an MLflow run identifier populated on every row.

The final six-column prediction set was written to workspace.student_aggregate.student_attrition_risk_prediction, linked to MLflow run 357d3a6420db4cfa80a626d5eff9ee20 and stamped scored_at 2026-08-06T15:46:31.493369Z. Retrieval by student_deidentified_hash was confirmed for the sampled students, demonstrating the operational output the application requires: a student-level risk score together with a clear at-risk or not-at-risk flag that can be retrieved from Delta by the student's de-identified key. Consistent with the limitations recorded in plan.md, attrition_risk_percentage must be presented to staff as a relative risk and ranking score rather than a calibrated likelihood of attrition, because class weighting recentres model scores around the 0.50 decision boundary. A stored score of 52 therefore indicates comparatively higher relative risk, not a 52 per cent chance that the student will attrit.

Source: Machine Learning Model(2).ipynb — Sections 21, 21.1 and Phase 3 prediction/persistence sections.

Generative AI Conversation and Reasoning summaries to generate documentation:

Generative AI declaration: Generative AI was used to assist in creating the evidence-pack template and documenting existing outputs from the Databricks machine learning notebook. The prompts used for this assistance are reproduced below for transparency. Model results and screenshots presented in this evidence pack were obtained from the implemented Databricks solution.

1. OpenAI prompt to generate template documentation:

Hello, create an editable Word template for an Evidence Pack regarding the Model Predictions section of a machine learning model. I have attached my notebook file detailing a random forest ML model to draw necessary information regarding model predictions.

The template should capture evidence from the trained model, including:

- validation and test performance metrics: Recall, Accuracy, Precision, F1, ROC-AUC, MSE, confusion-matrix counts and predicted-class totals.
- probability-distribution diagnostics and the non-triviality checks.
- final student-level attrition predictions persisted to workspace.student_aggregate.student_attrition_risk_prediction.
- the prediction output fields, 0.50 threshold logic, MLflow run traceability and scoring timestamp.

Structure this documentation with clear headings, tables for inserting metrics, sections to insert screenshot evidence of outputs, and explanations of the inserted information.

Ensure to add and discuss model limitations and design choices resulting in model predictions. Summarise model outputs demonstratively and provide me this template to be filled out as a .docx file.

2. Claude Code prompt to fill out template documentation:

Hello, you are currently on a shared databricks account webpage on a notebook file (ipynb). The associated .docx file has been attached, this is named: Model_Predictions_Evidence_Pack_Editable.docx it is found at the path "C:\Users\<local_user>\Downloads\Model_Predictions_Evidence_Pack_Editable.docx"

Changes in the bullet points must be made to the associated .docx file. In general, you can read through the file to understand the necessary tasks to complete, there are editable sections that must be filled out.

- Do not leave this page, do not investigate any other pages on databricks or access other webpages as you complete this task.
- At all costs you must not re-run the notebook file, do not run any codeblocks.
- Do not edit the code at all. Whether this is executable or commentary code, do not edit the code blocks.
- Look through the notebook files and retrieve these metrics:
- The outputs produced by the trained Random Forest Model, model performance metric outputs, probability diagnostics, and final student-level attrition predictions, insert a screenshot of the notebook outputs for these.
- Model Performance Metric Outputs: Recall, Accuracy, Precision, F1, ROC-AUC and MSE, confusion matrix counts and predicted class totals. Insert a screenshot of the output in the associated .docx file.
- Model performance metrics for validation and test regarding these metrics: Rows, Recall (primary), Accuracy, Precision, F1, ROC-AUC, MSE, True positives (TP), True negatives (TN), False positives (FP), False negatives (FN), predicted positive, predicted negative. Include these metric values in the associated .docx file.
- Probability Distribution Diagnostic outputs for validation and test splits as an inserted screenshot output.
- Include the output metrics of validation and test for: Minimum probability, Maximum probability, Mean probability, Probability standard deviation, Distinct probabilities, Predicted positive, Predicted negative. Insert a screenshot of non-triviality check also.
- Student Attrition Risk Predictions screenshot output, follow the recommended evidence insert for page 3.
- Complete the summarised evidence commentary section with your summarised evidence findings.

If you do not understand or require human input for completing a section, feel free to reach out and I will discuss further to complete this task.